TEST CODE **002492**

FORM TP 22138

MAY/JUNE 2002

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

PHYSICS

Paper 02 – General Proficiency

$1\frac{1}{2}$ hours

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. You MUST use this answer booklet when responding to the questions. For each question, write your answer in the space provided and return the answer booklet at the end of the examination.
2. ALL WORKING MUST BE SHOWN in this booklet, since marks will be awarded for correct steps in calculations.
3. Attempt ALL questions.
4. The use of non-programmable calculators is allowed.
5. Mathematical tables are provided.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

1. **You are to spend no more than $\frac{1}{2}$ hour on this question.**

A liquid, A, initially at temperature $T_1 = 28.4°C$, is heated by an immersion heater in a container. The resulting temperature, T_2, is recorded at 1 minute intervals and the following results obtained.

Time, t / min.	0	1	2	3	4	5	6	7
Temperature T_2/°C	28.4	36.0	43.1	52.5	60.5	69.0	75.1	84.3
Temperature changes ΔT/°C	0							

(a) Complete the table by computing the temperature changes $\Delta T = T_2 - T_1$.

(4 marks)

(b) Plot a graph of Temperature change, ΔT, against time, t, on page 3. **(9 marks)**

(c) Find the slope, S, of the graph.

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(5 marks)

(d) C_p the Specific Heat Capacity of the liquid is related to the slope S by:
$C_p = \frac{2000}{S}$ J kg^{-1} K^{-1}. Find the Specific Heat Capacity of the liquid.

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(2 marks)

(e) What would the temperature of the liquid after 12 minutes?

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(4 marks)

GO ON TO THE NEXT PAGE

(f) Another liquid, B, has a much higher specific heat capacity than Liquid A. On the same graph paper used in (b) on page 3 sketch **(DO NOT PLOT)** a line showing the approximate graph that would be obtained if Liquid B was the working fluid.

(2 marks)

(g) State TWO possible sources of error in this experiment and in EACH case describe a method that might be used to minimize the magnitude of the error.

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(4 marks)

Total 30 marks

2. (a) (i) Distinguish between density and relative density.

(2 marks)

(ii) Show why relative density has NO units.

(2 marks)

(b) (i) A block of wood measures 2 cm x 4 cm x 9 cm. Calculate its volume in SI units.

(2 marks)

(ii) Express the result in (b) (i) in standard form.

(1 mark)

(iii) If the mass of the block is 57.6×10^{-3} kg, calculate its density.

(2 marks)

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(c) (i) State Archimedes Principle.

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(1 mark)

(ii) Use Archimedes Principle to explain why some objects float while others sink in the same liquid.

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(2 marks)

(iii) Explain how it is possible to predict whether a given object will float or sink in a given liquid.

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(2 marks)

(d) Would the block of part (b) on page 5 float or sink in a liquid of density 810 kg m^{-3}?

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(1 mark)

Total 15 marks

3. (a) Calculate the resistance of an electrical heating coil rated at 275 W, 110 V.

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(4 marks)

(b) Two such coils are connected together in the circuit of a hair dryer by means of switches as shown in Figure 1.

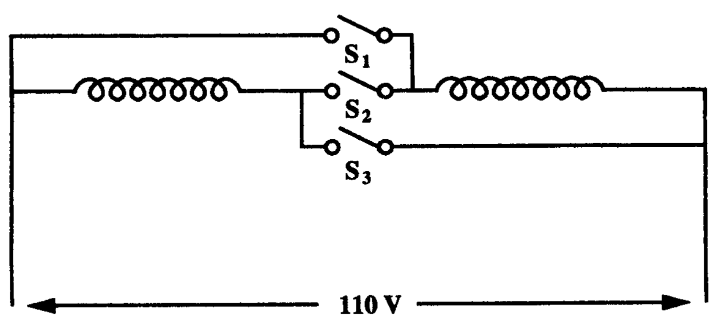

Figure 1

This dryer has two power settings LOW and HIGH. A setting is selected by opening or closing the switches. The table below shows the correspondence between power setting and the states of the switches.

POWER SETTING	S1	S2	S3
LOW	OPEN	CLOSED	OPEN
HIGH	CLOSED	OPEN	CLOSED

Draw the equivalent circuit for EACH setting by redrawing the circuit without the switches. In your equivalent circuit, open switches are to be left out entirely and closed switches are to be replaced by an unbroken line.

(2 marks)

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(c) Identify the type of connection employed in EACH case.

(2 marks)

(d) Compute the effective resistance of EACH connection.

(3 marks)

(e) Write down TWO characteristics of a fuse suitable for use in a circuit such as that of the hair dryer.

(2 marks)

(f) Draw a simple sketch showing how a fuse should be connected in a circuit to protect a load. In addition to the fuse and load your circuit should include: live and neutral wires, and an on/off switch.

(3 marks)

Total 16 marks

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4. (a) Describe an experiment to show that the pressure in a liquid increases with depth.

(4 marks)

(b)

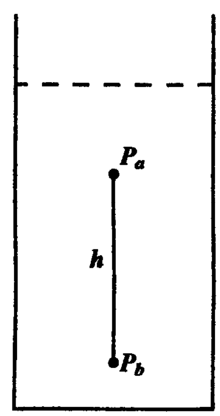

Figure 2

A container is filled with a liquid of density ρ as shown in Figure 2. Express the pressure difference between $P_b - P_a$ in terms of ρ and the height h.

(1 mark)

(c)

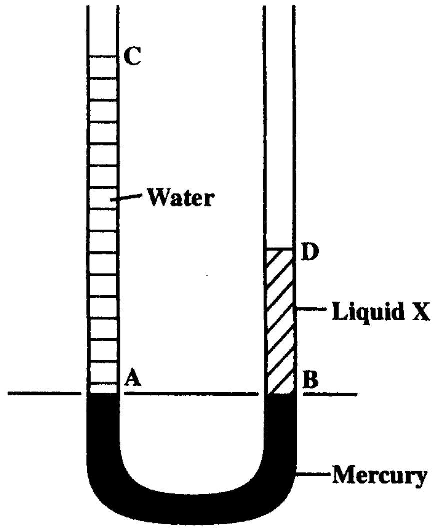

Figure 3

The apparatus shown in Figure 3 may be used to determine the density of an unknown liquid X.

The tube has a uniform cross-section and the height of the water column is twice that of liquid X. A and B are at the same horizontal level. The pressure difference between points A and C is 2.5 x 10^3 Pa.

(i) Calculate the height of the water column.

(Density of water = 1000 kg m^{-3}; g = 10 m sec^{-2}) **(3 marks)**

(ii) Express $P_B - P_D$ in terms of P_A and P_C. [Hint: the answer can be deduced without resorting to any further calculation.]

(2 marks)

(iii) What is the relationship between P_C and P_D?

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(1 mark)

(iv) Determine the density of liquid X.

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(4 marks)

Total 15 marks

5. (a) (i) Draw a diagram showing the structure of the carbon atom $^{12}_{6}C$.

(5 marks)

(ii) Carbon –14 is often referred to as an isotope of carbon. Explain what is meant by the term 'isotope'.

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(1 mark)

(b) Define 'half-life' when used in reference to a radio-active nuclide.

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(2 marks)

(c) A radioactive tracer has a half-life of 18 hours. Calculate the fraction of the initial activity remaining after 90 hours.

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(3 marks)

(d) In a radioactive process the element Radium (Ra) emits gamma radiation and decays to Radon (Rn) and Helium (He). The equation of this process is $^{226}_{88}Ra \longrightarrow ^{A}_{86}Rn + ^{4}_{z}He$. Determine the values of A and Z.

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(4 marks)

Total 15 marks

END OF TEST